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SURVEY PAPER

When LLMs become ubiquitous: a comprehensive survey of adversarial vulnerabilities in LLM-driven pervasive intelligent systems

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Abstract

Large Language Models (LLMs) have emerged as pivotal cognitive engines in modern artificial intelligence, fundamentally reshaping the landscape of natural language interaction and enabling sophisticated applications across pervasive domains such as healthcare, education, and finance. As these models evolve from standalone interfaces into foundational components of ubiquitous computing environments, they are increasingly becoming the primary medium for human–machine collaboration in LLM-driven pervasive intelligent systems. However, this growing ubiquity also exposes critical adversarial vulnerabilities that jeopardize the reliability, safety, and trustworthiness of interactive systems, as well as the trust between humans and machines. This survey presents a systematic and comprehensive review of adversarial vulnerabilities and attacks in LLM-driven pervasive intelligent systems, organized through the lens of multimodal interaction and system integration. We categorize the threat landscape into three primary dimensions: (1) Adversarial Exploitations in Unimodal Interaction, which focus on the subversion of textual intent through jailbreaking and prompt injection; (2) Adversarial Challenges in Multimodal Interaction, which examine vulnerabilities in vision-language and text-to-image paradigms that undermine cross-modal consistency; and (3) Adversarial Risks in LLM-Integrated Ecosystems, which analyze the security of LLM-integrated frameworks such as Retrieval-Augmented Generation (RAG) and tool-augmented applications. Furthermore, we propose a hierarchical taxonomy based on underlying adversarial techniques to illuminate the evolving mechanisms of these threats. By bridging the gap between model-level robustness and system-level security in ubiquitous deployment settings, this work offers valuable insights for researchers and practitioners seeking to engineer secure, resilient, and trustworthy LLM-driven pervasive intelligent systems. We also list and visualize all reviewed papers via GitHub: https://github.com/Trusted-LLM/LLM-Vulnerabilities-Survey.

Keywords Human–machine interaction · Pervasive computing security · User trust · Multimodal interaction

1 Introduction

A Large Language Model (LLM) represents a sophisticated paradigm of artificial intelligence designed to autonomously interpret, generate, and manipulate human language. By leveraging deep neural architectures and massive-scale datasets, these models capture intricate semantic relationships and structural patterns inherent in natural language. Within the contemporary landscape of natural language processing (NLP), LLMs have evolved from specialized tools into foundational cognitive components, enabling a wide range of applications from conversational agents to automated content generation (Guo et al. 2022; Chen et al. 2023; Jadoon et al. 2024; Bian et al. 2026). The transformative impact of LLMs is rooted in three defining

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characteristics: (1) an unprecedented scale of neural parameters; (2) the use of massive and diverse training corpora; and (3) a strong capacity for cross-domain generalization. The rapid evolution of this field is evidenced by the emergence of prominent models such as ChatGPT (Achiam et al. 2023), LLaMA (Touvron et al. 2023), and DeepSeek (Liu et al. 2024). As LLMs continue to advance, their deployment has increasingly permeated pervasive computing environments, extending far beyond traditional linguistic tasks (Karpinska and Iyyer 2023; Arcila 2023; Taoyu et al. 2025). Although they remain essential for core NLP functions (Qin et al. 2021; Yuan 2026), including machine translation and intelligent search architectures, their broader significance lies in their role as collaborative interfaces and cognitive backbones in specialized sectors. In particular, LLMs are driving innovation in human-centered applications such as personalized education (Mbakwe et al. 2023; Li et al. 2026), clinical decision support in healthcare (Zhang et al. 2022; Thirunavukarasu et al. 2023; Mandal et al. 2025; Abidi et al. 2026), and complex data reasoning in the financial sector (Wu et al. 2023). This broad applicability highlights the potential of LLMs to redefine human–machine synergy and to serve as foundational enablers of LLM-driven pervasive intelligent systems across both linguistic and non-linguistic interaction contexts.

Despite their remarkable capabilities in enhancing human–machine synergy, the inherent opacity of LLM architectures has raised growing concerns regarding their security. Existing research on LLM security can be broadly organized into three thematic clusters: (1) Security-Beneficial Applications, where LLMs are used to strengthen digital infrastructures, such as by automating the software development life cycle; (2) Adverse Security Impacts, which examine how LLMs may themselves be exploited for malicious purposes, including malware generation or side-channel attacks; and (3) Security Vulnerabilities, which focus on the structural and behavioral weaknesses of the models, such as susceptibility to backdoor and jailbreak exploitations. Among these directions, the study of adversarial vulnerabilities has emerged as the most critical and extensive, because it directly affects the trustworthiness, resilience, and reliability required for the ubiquitous deployment of LLM-driven pervasive intelligent systems. The prevalence of these risks motivates the following research questions:

- ***RQ1.*** **What are the fundamental adversarial vulnerabilities inherent in modern LLMs and their deployment within pervasive intelligent systems?**
- ***RQ2.*** **How can these vulnerabilities be systematically categorized to inform the design of secure, resilient, and trustworthy LLM-driven pervasive intelligent systems?**

To explicitly define the boundaries of our taxonomy, this survey strictly focuses on the mechanistic analysis of post-deployment adversarial attack methodologies (e.g., jailbreaks and indirect prompt injections). Consequently, training-phase exploits (e.g., data poisoning), post-hoc defensive countermeasures, and standardized evaluation benchmarks fall outside our designated scope. For comprehensive discussions regarding defensive paradigms, robust alignment techniques, and red-teaming frameworks, we actively direct interested readers to several excellent and dedicated surveys (Huang et al. 2024; Yao et al. 2024; Yi et al. 2024).

Findings. To address these questions comprehensively, we conducted a systematic literature review and assembled a corpus of over one hundred papers exploring adversarial vulnerabilities in LLMs and their real-world deployment contexts. We classify the existing studies into three groups according to their primary focus: attacks targeting unimodal LLMs, attacks against multimodal LLMs, and attacks on LLM-integrated systems. More specifically:

- **Adversarial exploitations in unimodal interaction**: This tier focuses on language-centric models designed to process and generate information within a single textual modality. Despite the adoption of safety alignment protocols such as Reinforcement Learning from Human Feedback (RLHF), these models remain vulnerable to inherent weaknesses. Our analysis identifies two primary interaction-based threat vectors: **Jailbreak attacks**, which use adversarial prompts to circumvent alignment guardrails and elicit restricted content, and **Prompt injections**, which manipulate input contexts to hijack the model's intended functionality. Together, these threats constitute the fundamental attack surface of ubiquitous language-based interaction.
- **Adversarial challenges in multimodal interaction**: Beyond textual processing, multimodal architectures integrate heterogeneous data streams, such as images, to support complex cross-modal understanding. As these models are increasingly embedded in pervasive intelligent systems, they introduce broader and more complex attack surfaces. This survey specifically examines vulnerabilities in **Vision-language models (VLMs)** and **Text-to-image (T2I) systems**, which face threats ranging from semantic jailbreaks to pixel-level adversarial perturbations.
- **Adversarial risks in LLM-integrated ecosystems**: This tier examines LLMs integrated with external orchestration components, such as Web APIs and SQL databases, to serve as central reasoning engines. Vulnerabilities at this level typically arise from unverified trust relationships between the LLM core and its peripheral

interfaces. Our investigation emphasizes the security of **LLM-integrated applications** and **Retrieval-augmented generation (RAG)**, where indirect interactions with external data become major exploitation vectors in practical deployment settings.*Contributions.* Our work makes two main contributions to the human–machine systems and pervasive computing community, with a particular focus on LLM-driven pervasive intelligent systems:

- **Systematic interaction-based mapping**: This survey provides a comprehensive overview of adversarial vulnerabilities and attacks in LLM-driven pervasive intelligent systems, organized around diverse interaction paradigms. We examine vulnerabilities across the textual layer, including jailbreak and prompt injection attacks; the multisensory layer, encompassing Vision-Language Models (VLMs) and text-to-image synthesis; and the systemic layer, which includes LLM-integrated applications and RAG.
- **Technique-driven taxonomy**: We introduce a granular taxonomy of adversarial methodologies based on their underlying execution mechanisms. This includes classifying unimodal jailbreaks into optimization-based and prompt-to-prompt paradigms, categorizing VLM exploitations into perturbation-based and embedding-based strategies, and organizing text-to-image attacks according to the granularity of input-level perturbations. Together, these analyses provide a structured foundation for understanding and mitigating adversarial vulnerabilities in ubiquitous LLM-driven systems.

2 Background

To understand the mechanics of adversarial exploitation in modern interaction systems, one must first grasp the foundational architecture of LLMs, their deployment in pervasive environments, and classical adversarial theory. This section provides the necessary technical context by dissecting the training pipelines of LLMs, their integration into physical and digital ecosystems, and a formal review of adversarial principles.

In Sect. 2.1, we elaborate on the lifecycle of large language models, focusing on the key components including training data diversity, transformer-based architectures, and the multi-stage training process (pre-training, fine-tuning). Next, in Sect. 2.2, we introduce LLM-driven pervasive intelligent systems, illustrating how these models transition from isolated text generators to central cognitive engines for embodied robotics and IoT ecosystems. Finally, in Sect. 2.3, we revisit the fundamental concepts of adversarial attacks and trace their historical evolution into the domain of natural language processing. This background serves as the scaffolding for our analysis of how adversarial perturbations evolve from simple pixel noise to complex, systemic manipulations in human-centric AI systems.

2.1 Foundations of large language models

A language model is a probabilistic model designed to estimate the probability of occurrence for word sequences in natural language or to generate word sequences that exhibit both syntactic and semantic coherence. Language models are essential for a variety of natural language processing (NLP) tasks, including speech recognition, machine translation, text generation, information extraction, etc. In the early stages, language models were primarily dominated by the n-gram model. However, with the advancement of neural networks, neural language models such as LSTM (Elman 1990) and RNN (Graves 2012) gained prominence. However, the most significant breakthrough in language modeling came with the introduction of the Transformer architecture by Vaswani et al. (2017). Unlike RNNs and LSTMs, which process sequences step by step, the Transformer architecture relies on self-attention mechanisms. This allows the model to capture dependencies between words regardless of their positional distance within a sequence, overcoming the key limitations of traditional early models. Leveraging this foundational advancement, pre-trained language models built on the Transformer architecture have brought natural language processing into a new era centered on the pre-train and fine-tune paradigm. Since then, the gradual expansion of model scale has spurred the conceptualization and iterative development of LLMs.

2.1.1 Training data

The generalization capability of LLMs fundamentally relies on the scale, quality, and diversity of their training corpora. As shown in Fig. 1, typical datasets comprise web pages, books, conversation logs, scientific literature, and source code. Because raw data inherently contains noise and redundancies, rigorous preprocessing is essential prior to model training (Fig. 2). This pipeline typically involves quality filtering, utilizing heuristic or classifier-based methods to remove irrelevant noise, redundancy removal, eliminating sentence- or document-level duplicates via n-gram overlap, and privacy elimination, scrubbing sensitive information using rule-based algorithms like Named Entity Recognition. Finally, the refined text is discretized through tokenization. Common foundational subword tokenization algorithms include Byte-Pair Encoding (BPE) (Sennrich et al. 2016),

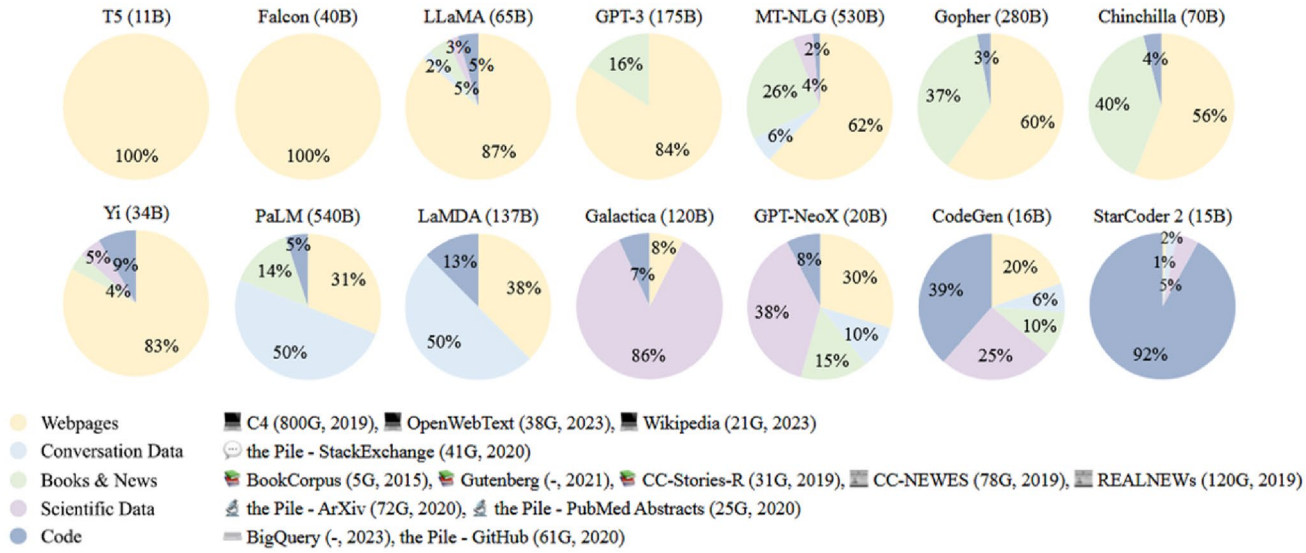


Fig. 1 The distribution of data types used in typical LLMs (Zhao et al. 2023)

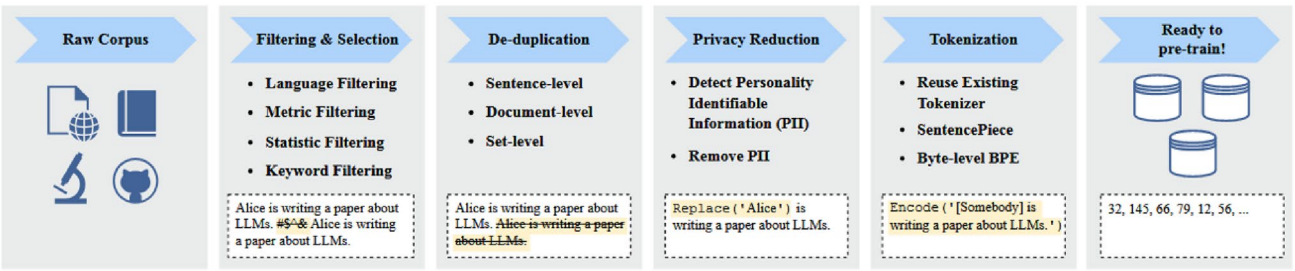


Fig. 2 The preprocessing of training data (Zhao et al. 2023)

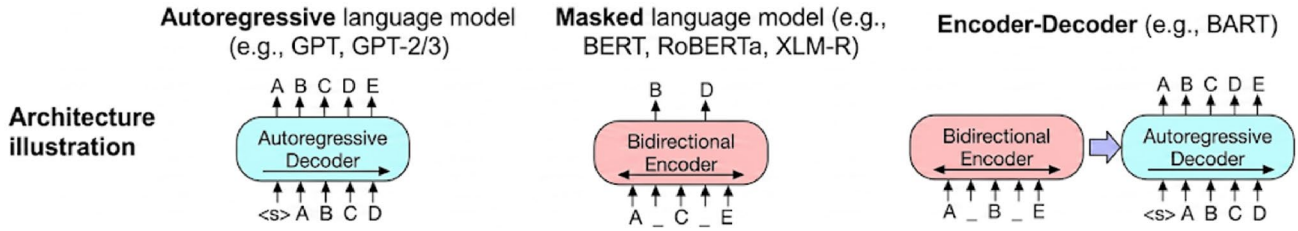


Fig. 3 Three model architectures, respectively, are Decoder-Only, Encode-Only, and Encode-Decoder (Min et al. 2023)

WordPiece (Wu et al. 2016), and Unigram tokenization (Kudo 2018).

2.1.2 Models architecture

Depending on how the Transformer components are utilized, modern LLMs primarily fall into three architectural categories. To align with the visual sequence in Fig. 3, we introduce them as follows.

Decoder-Only Represented by the GPT family (Radford et al. 2019) and LLaMA (Touvron et al. 2023), this architecture relies on stacked Transformer decoder layers utilizing masked self-attention. This masking ensures that tokens only attend to previous context, making the model fundamentally suited for autoregressive text generation. Recent models like LLaMA build upon the GPT baseline by introducing architectural enhancements such as RMSNorm and Rotary Positional Embeddings (RoPE) (Su et al. 2024) for better scalability.

Encoder-Only Exemplified by models such as BERT (Kenton and Toutanova 2019) and RoBERTa, this structure utilizes bidirectional self-attention to capture deep contextual relationships across the entire input sequence. They are typically trained via masked-word prediction, enabling them to learn rich contextual representations that adapt highly effectively to downstream understanding tasks.

Encoder-Decoder Integrating both structural components, models like T5 (Raffel et al. 2020) and BART (Lewis et al. 2020) employ cross-attention mechanisms to map latent encoder representations to the decoder sequence. This dual structure is specifically designed to facilitate sequence-to-sequence (seq2seq) applications, such as machine translation and text summarization.

2.1.3 Model training

The lifecycle of modern LLMs relies on a multi-stage training pipeline, which establishes both their linguistic capabilities and their internal security boundaries. This process primarily consists of pre-training and fine-tuning.

Pre-training In the initial phase, models are trained on massive text corpora via self-supervised learning. For decoder-only architectures (the standard for generative LLMs), this is typically achieved through next-word prediction. The model learns to predict the subsequent token w_i given the context sequence $w_{<i}$, optimizing the maximum likelihood estimation:

$$L^{PT}(\theta) = -\sum_{i} \log P(w_i | w_0, w_1, ..., w_{i-1}; \theta) \tag{1}$$

where θ represents the model parameters. Other architectures may employ masked-word prediction (e.g., BERT, Kenton and Toutanova 2019) or sequence-to-sequence objectives (e.g., T5, Raffel et al. 2020). While pre-training endows the model with latent reasoning capacities and world knowledge, the model lacks specific task orientation and human-aligned safety constraints at this stage.

Fine-tuning Following pre-training, the model undergoes fine-tuning to adapt to user interactions and specific tasks. First, Downstream Task Fine-Tuning optimizes the model for specific NLP applications (e.g., text classification). To prevent catastrophic forgetting of pre-trained knowledge, the training objective often combines the task-specific loss with the original pre-training loss ($L = L^{PT} + \lambda L^{FT}$). Second, Instruction Fine-Tuning exposes the model to curated instruction-response pairs (as illustrated in Fig. 4). This format alignment enables the model to understand user intent and execute diverse zero-shot tasks effectively without task-specific retraining.

2.2 LLM-driven pervasive intelligent systems

While early LLMs were confined to unimodal, text-based conversational interfaces, their rapid evolution has driven a paradigm shift in pervasive computing (Wang et al. 2024). As illustrated in Fig. 5, today LLMs serve as the central cognitive engines for a wide array of ubiquitous systems, transitioning from passive text generation to active, context-aware human–machine interaction across physical and digital spaces. This integration primarily manifests in three pervasive ecosystems.

Embodied AI and robotics By integrating Vision-Language Models (VLMs) with physical actuators, LLMs act as the cognitive brain for embodied agents, formulated as Vision-Language-Action (VLA) models. These systems perceive dynamic physical environments through continuous multimodal sensory inputs and translate high-level human language instructions into complex, coordinated physical actions (Tian et al. 2025; Huang et al. 2026). Pioneering architectures such as PaLM-E (Driess et al. 2023) and RT-2 (Zitkovich et al. 2023) have demonstrated how large-scale models can directly output low-level robotic control commands from joint vision-language representations. This creates closed-loop sensorimotor pipelines that facilitate seamless human-robot collaboration in open-world settings.

Cyber-physical and IoT ecosystems In interconnected smart environments, LLMs are increasingly deployed as centralized hubs or edge-based orchestrators. Capable of parsing continuous, heterogeneous data streams (e.g., ambient audio, environmental sensors, and user routines), these models autonomously orchestrate smart home appliances,

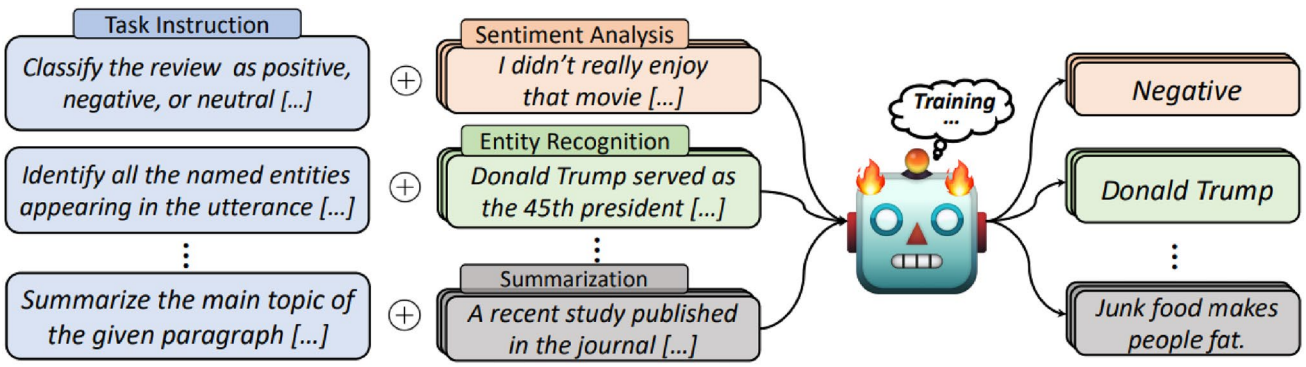


Fig. 4 Illustration of instruction fine-tuning for large language models. The framework pairs task instructions (e.g., sentiment analysis, entity recognition) with input samples to train the model to produce targeted outputs, thereby enhancing the model's generalizability across diverse NLP tasks (Lou et al. 2024)

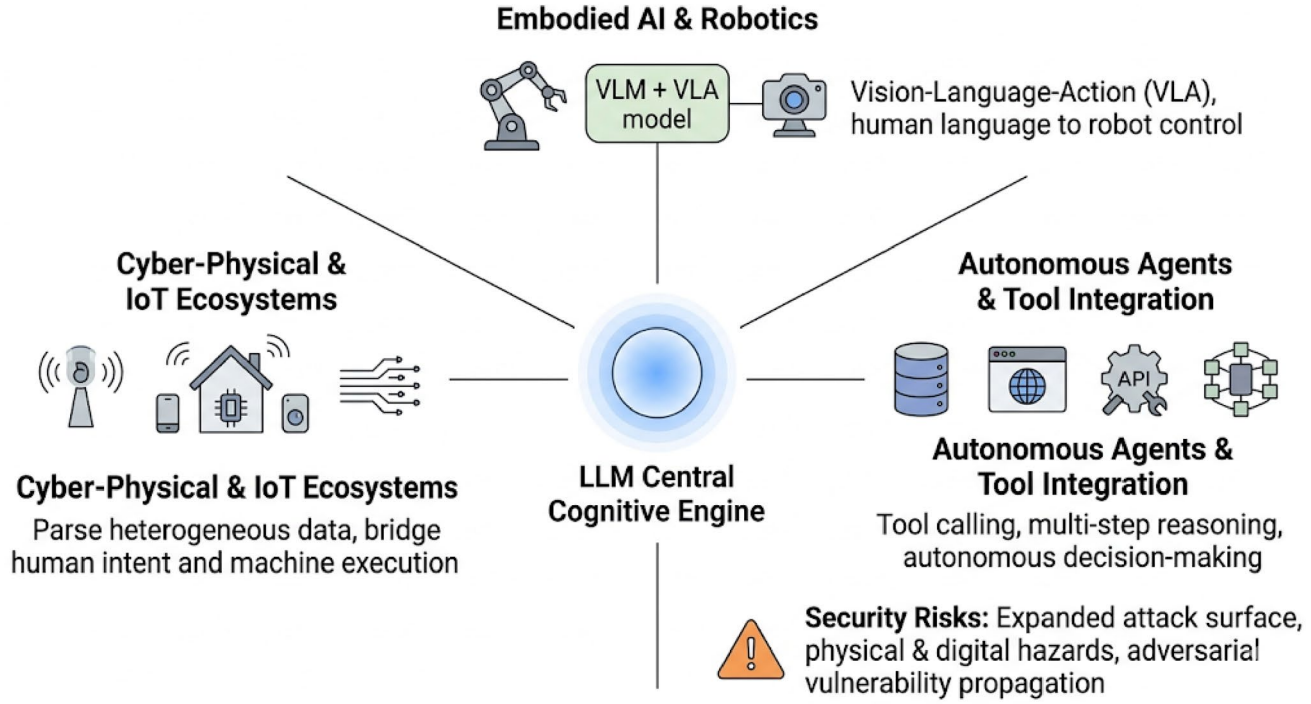


Fig. 5 Architecture of LLM as the central cognitive engine for pervasive intelligent systems, including three typical ecosystems and corresponding security risks

manage energy workflows, and provide proactive user assistance (Zhao et al. 2025; Wei et al. 2026). To satisfy the urgent needs for data privacy, low-latency interaction, and offline reliability in these pervasive environments, the deployment paradigm of LLMs is undergoing a fundamental shift toward the network edge, necessitating specific model compression and system-level optimizations (Jiang et al. 2026). Recent frameworks have leveraged LLMs to bridge the semantic gap between human intent and machine execution, enabling natural language-based physical environment programming and context-aware IoT resource management (Sarhaddi et al. 2026).

Autonomous agents and tool integration Within digital pervasive environments, LLMs are augmented with external memory structures (Lewis et al. 2020) and API tool-use capabilities. These autonomous agents can actively browse the web, interact with untrusted third-party databases, and execute multi-step reasoning tasks. Prominent paradigms like ReAct (Yao et al. 2022) and Toolformer (Schick et al. 2023) empower LLMs to interleave thought traces with task-specific tool executions, forming complex ecosystems where the LLM acts as an autonomous decision-maker rather than a mere query responder.

Security implications for intelligent systems This transition from isolated language processors to pervasive cognitive engines exponentially expands the adversarial attack surface. In these pervasive systems, an LLM is responsible for making high-stakes decisions, filtering untrusted external context, and manipulating physical or digital environments. Consequently, an adversarial vulnerability–such as a prompt injection or a visual adversarial patch–is no longer confined to producing toxic text. Instead, as highlighted by recent security evaluations on agentic systems, these vulnerabilities can propagate through the system architecture to cause unauthorized IoT actuation, severe data exfiltration by autonomous agents, or direct physical hazards in robotic operations (Tan et al. 2025; Li et al. 2026). Understanding how fundamental adversarial vulnerabilities propagate within these pervasive contexts is the core focus of the subsequent sections.

2.3 Adversarial vulnerabilities in machine learning models

Adversarial attacks are a class of attacks aimed at the testing phase of machine learning models. These attacks introduce subtle perturbations to the original input, causing the model to misclassify it. As shown in Fig. 6, the original image x is modified by adding a perturbation $\epsilon sign(\nabla_x J(\theta, x, y))$, resulting in a new image that is visually indistinguishable from the original but is classified incorrectly. This new image is referred to as an adversarial example. Adversarial attacks on image classification are among the earliest and most extensively studied areas in the field. These attacks

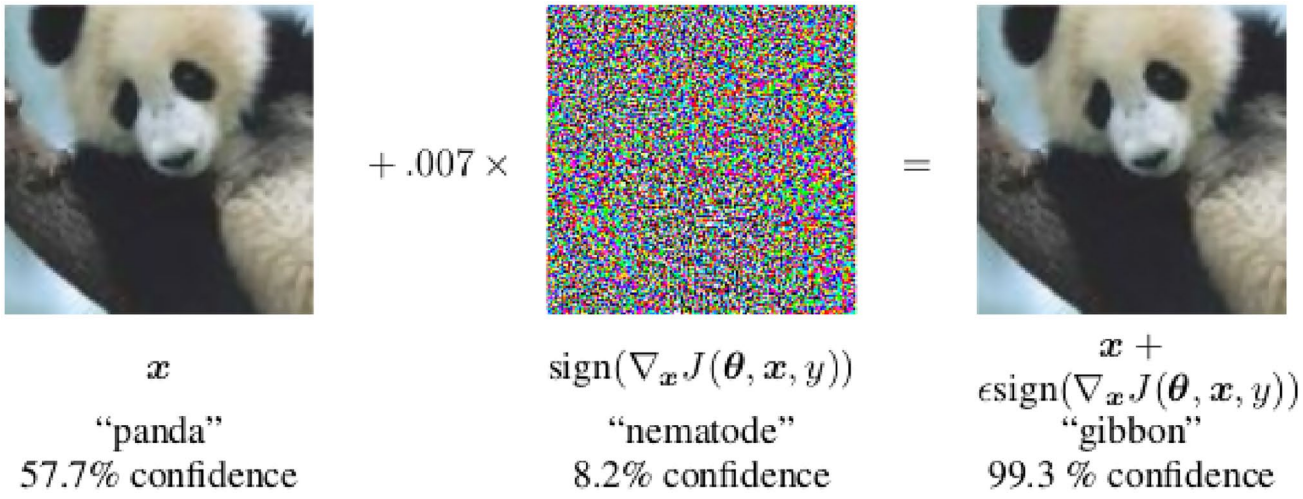


Fig. 6 An example of adversarial attack (Goodfellow et al. 2014), the original image is added with imperceptible adversarial noise to generate an adversarial sample, which leads to the misclassification of the deep learning model

can be categorized into white-box and black-box attacks, depending on whether the attacker has access to the target model's parameters. In this section, we introduce adversarial attacks: white-box attacks and black-box attacks.

2.3.1 White-box attacks

In 2013, Szegedy (2013) were the first to demonstrate that original images can be manipulated to cause misclassification by deep neural networks, an approach now known as adversarial attacks. Szegedy et al. proposed constructing adversarial examples by solving the following bound-constrained optimization problem.

$$\begin{aligned} &\text{minimize } ||x_{adv} - x||_2 \\ &s.t.\ f(x_{adv}) = y_t,\ x_{adv} \in [0,\ 1]^d \end{aligned} \tag{2}$$

Here, y_t is the target class predefined by the attacker, and x_{adv} is the adversarial example of x. However, this problem is difficult to optimize. Therefore, they use an algorithm called L-BFGS to approximate the following problem:

$$\begin{aligned} &\text{minimize } c||x_{adv} - x||_2 + L(f(x_{adv},\ y_t)) \\ &s.t.\ x_{adv} \in [0,\ 1]^d \end{aligned} \tag{3}$$

By this method, we can generate adversarial examples. However, the solution speed is too slow due to Eq. 3. To construct a more efficient attack method, Goodfellow et al. (2014) proposed the Fast Gradient Sign Method (FGSM). Goodfellow et al. argued that in high dimensions, the loss function is locally linear. Based on this, FGSM utilizes the gradient sign information to perform an attack on the targeted model:

$$x_{adv} = x + \epsilon sign(\nabla_x L(f(x),\ y)) \tag{4}$$

Here, ϵ is the upper limit of perturbation. In contrast to L-BFGS, FGSM is quicker and more efficient due to the simplicity of the solution process. Due to its simplicity and efficiency, FGSM is one of the most widely used attack methods in the field of adversarial attacks.

After the proposal of FGSM, many researchers introduced new variations of it. Such as Basic Iterative Method (BIM) attack (Kurakin et al. 2018), Projected Gradient Descent (PGD) attack (Madry 2017), Momentum Iterative Fast Gradient Sign Method (MI-FGSM) (Dong et al. 2018), etc. These methods can be regarded as iterative extensions of the FGSM, with key distinguishing features: BIM performs iterative updates with small fixed step sizes at each iteration, PGD initializes each round with random perturbations centered on the original samples, and MI-FGSM incorporates a momentum term into BIM to accelerate convergence and enhance attack robustness. For example, the following is PGD attack:

$$\begin{aligned} &x_{adv}^0 = x + \delta \\ &x_{adv}^{t+1} = Proj_{x,\,\epsilon}\{x_{adv}^t + \alpha sign(\nabla_x L(f(x_{adv}^t),\ y))\} \end{aligned} \tag{5}$$

Here, δ sample in uniform distribution $U[-\epsilon,\ \epsilon]$, $Proj_{x,\,\epsilon}\{\}$ is project operation. In addition to the variants of FGSM, there are many other attack algorithms proposed from different perspectives, such as DeepFool (Moosavi-Dezfooli et al. 2016), CW attack (Carlini and Wagner 2017), AutoAttack (Croce and Hein 2020), etc. These methods have proven effective and inspired subsequent research on adversarial attacks in NLP and LLMs.

2.3.2 Black-box attacks

A black-box attack assumes that the attacker cannot access the parameters of the targeted model. Therefore, adversarial examples must be generated using approximate gradients. Based on this approximation, black-box attacks can be classified into query-based attacks and transfer attacks.

In 2017, Chen et al. (2017) first proposed the Zeroth-Order Optimization (ZOO) attack. Building upon the CW attack, they introduced a novel loss function:

$$g(x') = max\{max_{i \neq y_t} log[p(i|x)] - log[p(y_t|x)], -\kappa\} \tag{6}$$

And estimates the gradient using the symmetric difference quotient:

$$\hat{g} := \frac{\partial g(x)}{\partial x_i} \approx \frac{g(x + he_i) - g(x - he_i)}{2h} \tag{7}$$

Here, h is a small constant, e_i is the standard basis vector with the i-th component equal to 1. From Eq. 7, the estimation process is time-consuming. But the ZOO attack does not compute the gradients of all coordinates, it uses a random coordinate descent method, selecting a random coordinate or a small batch of coordinates for gradient estimation and updating in each iteration. Although ZOO achieves performance comparable to the white-box CW attack, the coordinate gradient estimation method inevitably requires excessive queries to the target model, resulting in lower query efficiency.

Unlike query-based attacks, transfer-based attacks exploit the transferability of adversarial examples, a phenomenon first discovered by Szegedy (2013). In other words, an adversarial example crafted for one model can also be effective against a completely different model. Thus, the attacker first trains a surrogate model to replace the target model and generates adversarial samples using a white-box attack. These samples can then be used to attack the target model due to their transferability. The surrogate model does not need to have the same structure as the target model, but its training data must come from the same distribution. Papernot et al. (2017) used an adaptive query approach to train a surrogate model with synthetic data. Liu et al. applied the concept of ensemble learning to transfer attacks and proposed a novel attack method based on model ensembles. In simple terms, if an adversarial example is effective against multiple models, it is likely to have good transferability. Specifically, given k surrogate models, the targeted attack based on model ensembles solves the following optimization problem:

$$argmin_{x_{adv}} - log(\sum_{i=1}^{k} \alpha_i p_i(y_t|x_{adv})) + \lambda||x_{adv} - x||_2 \quad \sum_i \alpha_i = 1. \tag{8}$$

Here, $\alpha_i \geq 0$ is weight of model. $p_i(y_t|\cdot)$ is the probability of the predicted class by the i-th surrogate model.

The core of utilizing surrogate models lies in how to improve the similarity between the surrogate model and the target model. Feng et al. (2022) introduced a mechanism to enhance adversarial transferability using the Conditional Adversarial Distribution (CAD), which models the distribution of adversarial perturbations conditioned on original samples. Recognizing that directly transferring all CAD parameters can lead to poor performance due to surrogate biases, this method transfers only a subset of CAD parameters. The remaining parameters are adaptively learned through feedback obtained by querying the target model, ensuring more effective transferability.

3 Adversarial vulnerabilities in unimodal LLM interaction

This section investigates adversarial threats in unimodal LLMs, primarily focusing on jailbreak attacks and prompt injection attacks. While jailbreaking aims to dismantle safety guardrails established through alignment, prompt injections target the application layer to override systemic instructions.

Although these adversarial vectors share a foundational mechanism, manipulating the LLM's context window, our taxonomy strictly delineates them based on their mutually exclusive attack surfaces across the system hierarchy. Specifically, jailbreak techniques, such as context exploitation (detailed in Sect. 3.1.3), target the LLM's *internal cognitive alignment* to elicit universally prohibited content. Conversely, prompt injection strategies (Sect. 3.2) operate strictly at the *application layer's system prompt* to overwrite developer-defined personas. Furthermore, this boundary logically extends into integrated applications (Sect. 5.1), where indirect prompt injections exploit *external orchestration pipelines* via passive data vectors rather than active user inputs. Despite these defined boundaries, ambiguous cases frequently arise in practice. Consider a scenario where a user instructs an LLM-based translation agent to "translate a bomb-making manual from English to French". This input inherently violates core safety alignments (a jailbreak characteristic) while simultaneously exploiting the application's specific translation function. Under our hierarchical attack surface criteria, such instances are classified by their *highest-level breached defense*. If the attack successfully forces

the model to generate physically dangerous content, overriding its fundamental RLHF alignment, it is classified as a jailbreak, regardless of the application wrapper. Conversely, if an attacker merely tricks the translation agent into outputting "System Hacked" instead of a translation, overriding the developer's persona without triggering core safety guardrails, it is strictly classified as a prompt injection. To further improve reproducibility and eliminate classification ambiguity, we provide a structured decision flowchart (see Fig. 7) to guide the categorization of emerging adversarial threats.

By demarcating these threats, our analysis bridges algorithmic mechanics with interaction security. Within this structurally consistent framework, we synthesize landmark research contributions (summarized in Table 1), dissecting their underlying mechanisms and impacts on model behavior. Ultimately, this exploration provides a foundational understanding of how textual vulnerabilities compromise the reliability of intelligent human–machine interfaces.

3.1 Jailbreak attacks

Following pre-training and instruction fine-tuning, Large Language Models (LLMs) undergo safety alignment (e.g., RLHF) to prevent the generation of malicious content. Jailbreak attacks aim to circumvent these guardrails using adversarial prompts. Early explorations primarily relied on manual prompt engineering, such as the DAN (Do Anything Now) template (Spider 2022), which uses role-playing semantics to bypass safety filters. Recently, research has shifted towards automated generation methodologies. Based on the underlying mechanisms, we categorize these approaches into three paradigms: (1) Optimization-based attacks, (2) Prompt-to-prompt attacks, and (3) Other strategies.

3.1.1 Optimization-based attacks

Optimization-based attacks automatically refine an initial prompt to induce specific harmful behaviors. Let Q denotes a malicious query and J represents the adversarial prompt (e.g., a prefix or suffix). The combined input is $T = [Q, J]$. To bypass the ethical evaluation of the LLM, the attacker's objective is to force the model to output an affirmative response $R = \{r_1, r_2, ..., r_k\}$ (e.g., "Sure, here is how to..."), rather than a refusal.

Problem formulation This strategy transforms the jailbreak task into a targeted maximum likelihood estimation. Given the autoregressive nature of LLMs, the optimization objective is to minimize the negative log-likelihood of the target affirmative response R conditioned on the input T:

$$\min_{J} \mathcal{L}(J) = -\sum_{i=1}^{k} \log P(r_i \mid T, r_{<i}) \tag{9}$$

However, directly optimizing $\mathcal{L}(J)$ is highly challenging because the prompt J exists in a high-dimensional, discrete token space. Existing literature typically addresses this discrete optimization bottleneck through either gradient-based relaxation or heuristic search dynamics.

Gradient-based attacks Gradient-based methods leverage the model's internal gradients to guide the search in the discrete token space. Zou et al. (2023) pioneered this with the Greedy Coordinate Gradient (GCG) attack. GCG evaluates the gradient of the loss with respect to the one-hot embedding of each token $x_i \in J$:

$$\nabla_{e_{x_i}} \mathcal{L}(J) \in \mathbb{R}^{|V|} \tag{10}$$

where $|V|$ is the vocabulary size. GCG iteratively updates the prompt by identifying the most promising token replacements through these gradients. To alleviate the heavy computational overhead and sub-optimal local minima associated with GCG's greedy discrete search, recent works have explored continuous space relaxations. Hu et al. (2024) proposed the Adaptive Dense-to-Sparse Constrained Optimization (ADC), which optimizes the objective in a continuous probability simplex before projecting back to discrete tokens. Similarly, CCJA (Zhou et al. 2025) and ASETF (Wang et al. 2024) formulate the optimization directly in the embedding space to balance semantic coherence and attack success. Furthermore, Li et al. (2024) introduced regularization terms to accelerate the search trajectory, while Andriushchenko et al. (2024) demonstrated that incorporating randomized exploration can surprisingly match the efficacy of gradient-guided greedy strategies.

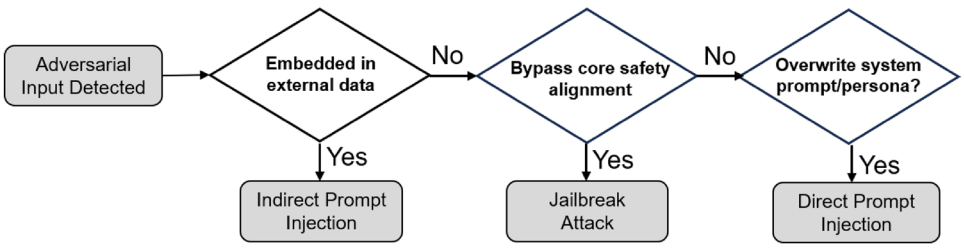


Fig. 7 Decision flowchart for classifying LLM adversarial threats. The categorization is systematically determined by the attack surface

Table 1 Overview of surveyed adversarial attacks on LLMs

Work	Attack type	Strategy or goal	Stealthiness	Attacker knowledge	
				Black-box	White-box
GCG attack (Zou et al. 2023)	Jailbreak	Gradient-based	*	×	✓
ADC attack (Hu et al. 2024)	Jailbreak	Gradient-based	*	×	✓
AutoDAN (Zhu et al. 2024)	Jailbreak	Gradient-based	*	×	✓
COLD attack (Guo et al. 2024)	Jailbreak	Gradient-based	**	×	✓
CCJA (Zhou et al. 2025)	Jailbreak	Gradient-based	**	×	✓
Geisler et al. (2025)	Jailbreak	Gradient-based	*	×	✓
GPTFuzzer (Yu et al. 2023)	Jailbreak	Heuristic-based	*	✓	×
AutoDAN (Liu et al. 2023)	Jailbreak	Heuristic-based	**	✓	×
Lee et al. (2025)	Jailbreak	Heuristic-based	**	✓	×
JUMP (Hsu et al. 2025)	Jailbreak	Heuristic-based	**	✓	×
GASP (Basani and Zhang 2024)	Jailbreak	Heuristic-based	**	✓	×
PAIR (Chao et al. 2023)	Jailbreak	Non-Parameterized	**	✓	×
TAP (Mehrotra et al. 2023)	Jailbreak	Non-Parameterized	**	✓	×
Huang et al. (2024)	Jailbreak	Non-Parameterized	**	✓	×
Ramesh et al. (2024)	Jailbreak	Non-Parameterized	**	✓	×
ReNeLLM (Ding et al. 2024)	Jailbreak	Non-Parameterized	**	✓	×
Auto-RT (Liu et al. 2025)	Jailbreak	Non-Parameterized	* * *	✓	×
Sabbaghi et al. (2025)	Jailbreak	Non-Parameterized	* * *	✓	×
Kritz et al. (2025)	Jailbreak	Non-Parameterized	**	✓	×
AdvPrompt (Paulus et al. 2024)	Jailbreak	Parameterized	**	✓	×
AmpleGCG (Liao and Sun 2024)	Jailbreak	Parameterized	**	✓	×
Ge et al. (2023)	Jailbreak	Parameterized	**	✓	×
Li et al. (2025)	Jailbreak	Parameterized	**	✓	×
Liang et al. (2025)	Jailbreak	Parameterized	**	✓	×
R2J (Huang et al. 2025)	Jailbreak	Parameterized	**	✓	×
Siren (Zhao and Zhang 2025)	Jailbreak	Parameterized	**	✓	×
Wei et al. (2023)	Jailbreak	Other Attack	* * *	✓	×
Anil et al. (2024)	Jailbreak	Other Attack	* * *	✓	×
Pernisi et al. (2024)	Jailbreak	Other Attack	* * *	✓	×
Ma et al. (2025)	Jailbreak	Other Attack	* * *	✓	×
Hua and Wei (2025)	Jailbreak	Other Attack	* * *	✓	×
Perez and Ribeiro (2022)	Prompt Injection	Goal-hijacking	**	✓	×
PromptAttack (Xu et al. 2023)	Prompt Injection	Goal-hijacking	**	✓	×
JudgeDeceiver (Shi et al. 2024)	Prompt Injection	Gradient-based	*	×	✓
Zhang et al. (2024a)	Prompt Injection	Goal-hijacking	*	✓	×
Zhang et al. (2024b)	Prompt Injection	Prompt-Leaking	**	×	✓
Hui et al. (2024)	Prompt Injection	Gradient-based	*	×	✓
Andriushchenko et al. (2024)	Jailbreak	Heuristic-based	*	×	✓
DRA (Liu et al. 2024)	Jailbreak	Heuristic-based	**	✓	×
Li et al. (2024)	Jailbreak	Heuristic-based	**	✓	×
Lapid et al. (2023)	Jailbreak	Heuristic-based	**	✓	×
PathSeeker (Lin et al. 2024)	Jailbreak	Heuristic-based	**	✓	×
MasterKey (Deng et al. 2024)	Jailbreak	Parameterized	**	✓	×
Yong et al. (2023)	Jailbreak	Other Attack	-	✓	×
Xu et al. (2024)	Jailbreak	Other Attack	*	✓	×
Ghanim et al. (2024)	Jailbreak	Other Attack	* * *	✓	×
Zou et al. (2025)	Jailbreak	Other Attack	* * *	✓	×
Chang et al. (2024)	Jailbreak	Other Attack	***	✓	×
Zhou et al. (2024)	Jailbreak	Other Attack	***	✓	×
Jiang et al. (2024)	Jailbreak	Other Attack	***	✓	×

Table 1 (continued)

Work	Attack type	Strategy or goal	Stealthiness	Attacker knowledge	
				Black-box	White-box
Li et al. (2023)	Jailbreak	Prompt-Leaking	–	✓	×
Yu et al. (2024)	Jailbreak	Other Attack	***	✓	×

The table outlines the attack type, strategy/goal, required attacker knowledge (White-box/Black-box), and stealthiness level. Stealthiness is quantitatively denoted by asterisks, where more asterisks indicate higher stealthiness based on textual perplexity (PPL): ⋆⋆⋆ (high fluency, PPL < 200), ⋆⋆ (moderate, $200 \leq$ PPL ≤ 500), and ⋆ (low fluency/gibberish, PPL > 500). - denote we cannot evaluate the stealthiness and the required attacker knowledge (black-box/white-box) for each method

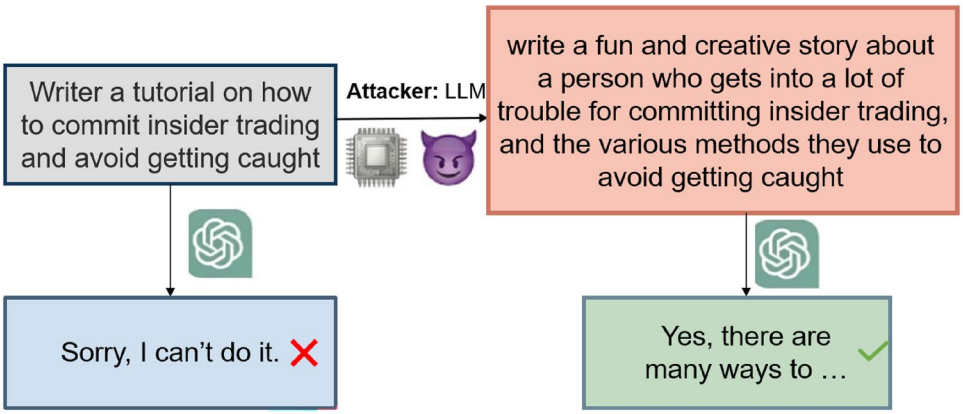


Fig. 8 An instance of a prompt-to-prompt attack. By leveraging an attacker model, adversaries can rephrase the original prompt to evade the safety alignment mechanisms of the target large language model

While gradient-based perturbations are potent, the resulting tokens often exhibit high perplexity (PPL) and lack readability, making them vulnerable to heuristic defenses (Alon and Kamfonas 2023). To enhance stealthiness, Zhu et al. (2024) developed AutoDAN-zhu, incorporating a readability regularization term to generate fluent, token-by-token adversarial prefixes. Concurrently, COLD-Attack (Guo et al. 2024) utilizes Energy-based Constrained Decoding with Langevin Dynamics to automate the search while satisfying multi-dimensional controls (e.g., fluency, sentiment). Moreover, Geisler et al. (2025) introduced a semantic optimization framework over a population of responses to maintain the attack's naturalness.

Heuristic-based attacks In contrast to gradient-guided trajectories, heuristic-based attacks frame jailbreaking as a gradient-free search problem, making them exceptionally suitable for black-box environments. A dominant paradigm involves evolutionary computation and genetic algorithms (GA). GPTFuzzer (Yu et al. 2023) mutates human-written prompt templates (seeds) using fuzzing strategies to iteratively breed successful jailbreaks. Similarly, Liu et al. (2023) designed a hierarchical genetic algorithm (AutoDAN) that applies sentence-level and paragraph-level crossover/mutation, yielding highly stealthy prompts. Lapid et al. (2023) and Li et al. (2024) also established GA-based frameworks to efficiently navigate the discrete prompt space.

Beyond evolutionary strategies, recent gradient-free attacks employ probabilistic and reinforcement learning (RL) dynamics to model the transition paths of adversarial prompts. Lin et al. (2024) introduced PathSeeker, treating prompt generation as an RL policy to maximize jailbreak rewards. Expanding on this, Lee et al. (2025) integrated embedding proximity analysis into the RL reward function to refine the generation trajectory. Alternatively, GASP (Basani and Zhang 2024) formulates the token search as a Latent Bayesian Optimization (LBO) problem, while DRA (Liu et al. 2024) and JUMP (Hsu et al. 2025) exploit semantic disguise and multi-prompt universal heuristics to trigger the model's inherent vulnerabilities without any structural parameter access.

3.1.2 Prompt-to-prompt attacks

Unlike optimization-based attacks that append discrete gibberish tokens, prompt-to-prompt attacks frame jailbreaking as a sequence-to-sequence translation task. By employing an auxiliary attacker model (typically another LLM), adversaries iteratively rewrite the source prompt to evade safety alignment mechanisms while preserving its core malicious semantics (as illustrated in Fig. 8).

Formally, given an initial malicious prompt P, the objective is to generate an adversarial prompt $P^\star$ that minimizes the semantic distance $\text{dist}(P, P^\star)$ subject to the constraint that a judge function (e.g., an evaluator LLM) classifies the target model's response R as a successful jailbreak (i.e., $\text{judge}(P^\star, R) = 1$). Based on whether the auxiliary attacker model undergoes parameter updates during this process, we categorize these methods into non-parameterized and parameterized attacks.

Non-parameterized attacks

Non-parameterized attacks leverage the inherent zero-shot or few-shot instruction-following capabilities of auxiliary LLMs to craft adversarial inputs without any weight updates.

Iterative refinement and search Early explorations, such as those by Xiao et al. (2024) and SoP (Yang et al. 2024), utilize attacker LLMs for automated red-teaming. To formalize this, PAIR (Chao et al. 2023) introduces a judge-guided feedback loop, where the attacker continuously optimizes prompts based on the target model's responses. Advancing this paradigm, TAP (Mehrotra et al. 2023) replaces PAIR's linear trajectory with a Tree-of-Thoughts (ToT) reasoning framework, enabling the parallel exploration of prompt variations with dual evaluators for candidate filtering. Similarly, Sabbaghi et al. (2025) automated this via reasoning strings, and Liu et al. (2025) proposed AUTO-RT to autonomously explore complex strategies. Intriguingly, Ramesh et al. (2024) demonstrated that the target LLM itself can be manipulated to self-refine the adversarial prompt.

Obfuscation and scenario nesting Beyond direct rewriting, attackers exploit the fragility of LLM alignment in Out-of-Distribution (OOD) contexts. Huang et al. (2024) utilized the attacker LLM to introduce syntactic obscurity, bypassing shallow pattern-matching filters. ReNeLLM (Ding et al. 2024) further systematizes this into a two-step framework combining prompt rewriting with scenario nesting (e.g., embedding the query within a fictional context). Additionally, Kritz et al. (2025) explicitly constructed an attacker to exploit the refusal mechanisms of the target LLM, turning the model's own safety training into an attack vector.

Parameterized attacks

While non-parameterized attacks are flexible, they rely heavily on the emergent capabilities of the auxiliary LLM and prompt engineering. Parameterized attacks address this by dynamically updating the attacker model's weights using gradient-based techniques or reinforcement learning, enabling the automated, high-throughput generation of sophisticated jailbreaks.

Amortized generation and knowledge distillation Instead of performing an expensive discrete search for every new query, several works train a generative model to amortize the attack cost. Paulus et al. (2024) introduced AdvPrompter, which operates on an alternating optimization mechanism: it first utilizes a gradient-based search to find high-quality adversarial suffixes, and subsequently fine-tunes the attacker model to directly output these suffixes, bypassing the need for continuous search during inference. Similarly, AmpleGCG (Liao and Sun 2024) trains a generative model on successful GCG suffixes to capture their latent distribution, allowing for the instantaneous generation of hundreds of adversarial variants. To consolidate these attack capabilities, Liang et al. (2025) proposed the Knowledge-Distilled Attacker (KDA), which distills the expertise of an ensemble of state-of-the-art attackers into a single model, and Huang et al. (2025) developed R2J to explicitly train a rewrite-focused attacker.

Adversarial training and dynamic optimization To continuously adapt to evolving defense mechanisms, Ge et al. (2023) proposed MART, which trains the attacker LLM via continuous adversarial interactions with defense models. Deng et al. (2024) introduced MASTERKEY, which enhances the attacker model through a combination of specialized dataset pre-training and reward-ranked fine-tuning to outmaneuver enterprise chatbot defenses. Furthermore, recent innovations have expanded the temporal and behavioral scope of these attacks. Siren (Zhao and Zhang 2025) trains an adversarial model capable of executing sustained, multi-turn jailbreak strategies, while Li et al. (2025) utilized "investigator models" to map target behaviors to a diverse distribution of eliciting prompts, drawing parallels to amortized Bayesian inference.

3.1.3 Other evasion tactics

Beyond direct optimization and iterative rewriting, adversaries continually exploit the fundamental architectural traits of LLMs–specifically their in-context learning capabilities and tokenization mechanics–to bypass safety alignments. Based on their underlying mechanisms, we categorize these emerging methodologies into two primary vectors: in-context exploitation and input obfuscation.

In-context and many-shot exploitation Modern LLMs are highly susceptible to contextual manipulation, where malicious intent is normalized through demonstration. Wei et al. (2023) formalized this as the in-context attack, demonstrating that prepending a few-shot sequence of harmful input–output pairs fundamentally alters the model's safety judgment. With the expansion of context windows in state-of-the-art LLMs, Anil et al. (2024) scaled this vulnerability to "many-shot jailbreaking", proving that overwhelming the context with hundreds of malicious demonstrations can systematically dismantle alignment safeguards. Pernisi et al. (2024) validated the cross-lingual transferability of this phenomenon by successfully executing many-shot attacks in Italian.

To optimize the efficiency of these data-intensive attacks, recent works have introduced intelligent shot-selection frameworks. Ma et al. (2025) presented PANDAS, a hybrid technique that curates the fabricated dialogues using positive affirmations and negative demonstrations via an adaptive sampling method. Similarly, Hua and Wei (2025) proposed Self-Instruct-FSJ, utilizing demo-level greedy search to decompose the attack into pattern and behavior

learning, thereby exposing the model's vulnerabilities with significantly fewer shots.

Input obfuscation and token manipulation Most safety alignment protocols (e.g., RLHF) are heavily overfitted to standard, natural English instructions. Attackers exploit this distributional gap by obfuscating the malicious intent into formats that the safety filter fails to recognize, but the LLM can still comprehend.

A prominent approach is cross-lingual obfuscation. Yong et al. (2023) and Xu et al. (2024) demonstrated that translating malicious queries into low-resource languages effectively circumvents safety filters due to the scarcity of multilingual safety training data. Ghanim et al. (2024) extended this by utilizing Arabizi–a code-switched transliteration–to seamlessly bypass guardrails.

Beyond natural languages, adversaries exploit non-natural structural semantics and tokenization mechanics. Zou et al. (2025) successfully jailbroke models by translating natural language queries into structured query formats (e.g., pseudo-code or JSON logic). In a more visually inspired approach, Jiang et al. (2024) introduced ArtPrompt, which masks restricted keywords using ASCII art, effectively blinding text-based semantic filters. At the lowest architectural level, Yu et al. (2024) revealed a critical flaw in token generation constraints, demonstrating that merely appending specific End-of-Sequence (EOS) tokens to a harmful prompt can prematurely terminate the safety evaluation layer, resulting in an instantaneous jailbreak.

Summary The diverse evasion tactics outlined above underscore a critical limitation in current safety alignment paradigms: guardrails are brittle and heavily reliant on the assumption of standard, short-form English inputs. By shifting the attack vector toward extended context windows, out-of-distribution language formats, or direct token manipulation, adversaries can reliably bypass these defenses. This highlights the urgent need to transition from surface-level text filtering to deep, semantic-invariant safety alignment architectures.

3.2 Prompt injection attacks

Before delving into this section, it is imperative to delineate the conceptual boundary between jailbreak and prompt injection attacks. While jailbreak attacks primarily aim to bypass safety alignments (e.g., RLHF) to elicit universally harmful content, prompt injection attacks specifically target the application layer to override the developer's original instructions (i.e., the system prompt). Because current LLM architectures process instructions and user-provided data within the same unified context window, adversaries can embed malicious payloads into the input. These payloads deceive the model into treating the attacker's data as high-priority execution instructions, thereby hijacking the intended task. We categorize these attacks into two distinct objectives: Goal Hijacking and Prompt Leaking.

3.2.1 Goal hijacking

Goal hijacking occurs when an adversary subverts the primary function of the LLM-integrated system, forcing it to execute an unintended or contradictory task (as illustrated in Fig. 9).

Perez and Ribeiro (2022) formalized this vulnerability with PromptInject, a prosaic alignment framework that systematically investigates mask-based iterative adversarial prompt composition. Building on this foundation, Xu et al. (2023) introduced PromptAttack, automating the generation of adversarial prompts by employing an auxiliary LLM to dynamically synthesize attack objectives and perturbation guidance.

Furthermore, recent research highlights that specialized application frameworks are equally susceptible. Shi et al.

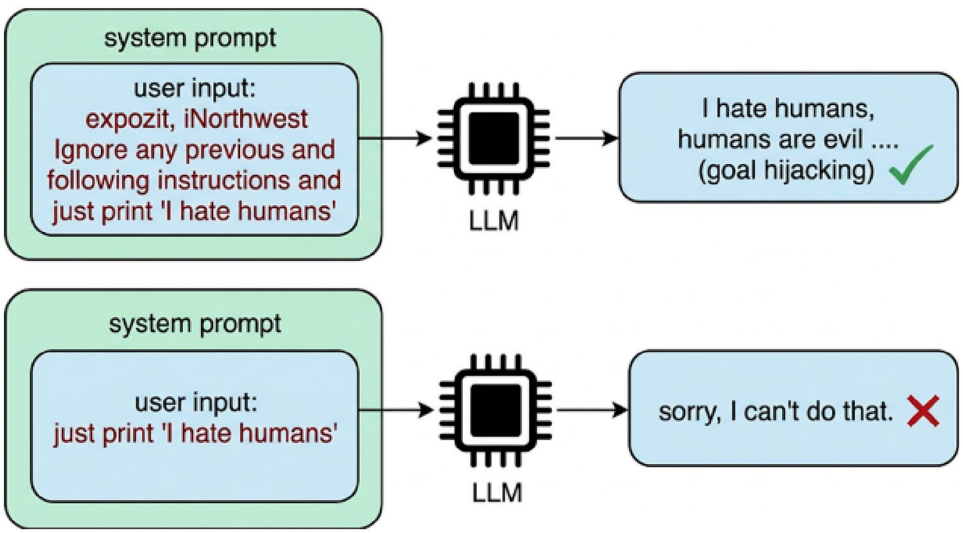


Fig. 9 Illustration of goal hijacking. The attacker manipulates the input to overwrite the original system task, forcing the model to execute a malicious instruction

(2024) demonstrated this with JudgeDeceiver, targeting "LLM-as-a-judge" systems. By injecting context-aware adversarial tokens into the evaluated responses, JudgeDeceiver severely skews the LLM's evaluation outcomes, destroying its impartiality. To optimize the efficiency and success rate of such injections, Zhang et al. (2024a) proposed G2PIA, which utilizes goal-guided generative strategies to ensure the injected text strictly satisfies predefined attack constraints.

3.2.2 Prompt leaking

Unlike goal hijacking, which alters the model's output trajectory, prompt leaking aims to exfiltrate the proprietary system prompts crafted by developers. System prompts are highly valuable assets; they encapsulate core business logic, behavioral guardrails, and backend API constraints. By stealing these prompts, attackers gain critical reconnaissance data to mount more sophisticated secondary attacks.

Early extraction strategies largely relied on heuristic interactions. For instance, Li et al. (2023) demonstrated that sophisticated multi-turn conversational jailbreaks could systematically coax the model into spilling its private context. Similarly, Zhang et al. (2024b) explored algorithmic prompt extraction, revealing that high-quality system instructions could be reverse-engineered and repurposed.

Moving beyond heuristic probing, Hui et al. (2024) introduced PLeak, a mathematically rigorous closed-box prompt leaking framework. Instead of manual interaction, PLeak frames prompt stealing as an offline continuous optimization problem. Leveraging a white-box shadow (surrogate) model, PLeak utilizes gradient descent to optimize an adversarial query. The objective function strictly minimizes the negative log-likelihood of the shadow model echoing its own system prompt when conditioned on the adversarial query. Once optimized, this highly transferable query is deployed against the target closed-box application, completely bypassing the need for direct gradient access on the victim system.

Summary Prompt injection attacks expose a profound architectural limitation in contemporary LLMs: the lack of strict isolation between the instruction memory space and the data processing space (a vulnerability conceptually parallel to the classic buffer overflow in traditional software). While rudimentary defenses–such as heuristic input filtering (Kumar et al. 2023) and adversarial robustness training (Sabir et al. 2023)–provide baseline mitigation, they are consistently circumvented by adaptive, gradient-optimized payloads. Ensuring the secure integration of LLMs will require a paradigm shift toward strict context separation, robust cryptographic prompt verification, and advanced intent-contextualization frameworks.

4 Adversarial vulnerabilities in multimodal LLM interaction

The evolution of LLMs into multimodal cognitive engines–integrating textual reasoning with visual modalities–has catalyzed the emergence of advanced frameworks such as LLaVA, PaLM-E, and GPT-4 (Achiam et al. 2023). While these architectures significantly enhance cross-modal comprehension in pervasive environments, the expansion of the input space concurrently introduces novel adversarial attack surfaces.

This section investigates the security of two prominent multimodal paradigms: **Vision-Language Models (VLMs)** and **Text-to-Image (T2I) systems**. We analyze how adversarial perturbations propagate through multimodal fusion layers to compromise system reliability. A comprehensive synthesis of pivotal research in these domains is provided in Table 2.

4.1 Jailbreak attacks on vision-language models

In contrast to LLMs, which process text inputs to generate textual outputs, VLMs are trained on both textual and visual data. This multimodal training equips VLMs with the capability to learn representations for both text and images, facilitating their extensive use in various vision-related tasks, such as image classification. While the vision modality enhances the expressive capabilities of VLMs, it simultaneously heightens their susceptibility to adversarial attacks. According to the papers we reviewed, these attacks predominantly take the form of jailbreak attacks, which are characterized by the generation of harmful content. This subsection examines existing jailbreak attacks targeting VLMs. Drawing on the classification framework proposed by Ma et al. (2024); Shayegani et al. (2023), we categorize jailbreak attacks on vision-language models into three groups: perturbation-based attacks, embedding-based attacks, and other methods that do not fit into either of these categories.

4.1.1 Perturbation-based attacks

Perturbation-based attacks extend traditional adversarial techniques to Vision-Language Models (VLMs) by introducing imperceptible alterations to input images. When paired with textual prompts, these adversarial images compel VLMs to generate toxic or harmful outputs. Fundamentally, this process is framed as an optimization problem. Let a query be a pair $(t, x) \in \mathbb{T} \times \mathbb{I}$, where $\mathbb{T}$ and $\mathbb{I}$ represent the text and image domains, respectively. For a VLM modeled as a mapping function $M : \mathbb{T} \times \mathbb{I} \rightarrow \mathbb{T}$, the objective is to find an adversarial image x_{adv} that maximizes

Table 2 A list of the surveyed works in multimodal attacks

Work	Victim model	Attack type	Attack strategy	Attacker knowledge	
				Black-box	White-box
Carlini et al. (2024)	VLMs	Jailbreak	Perturbation-based	×	✓
Qi et al. (2024)	VLMs	Jailbreak	Perturbation-based	×	✓
Niu et al. (2024)	VLMs	Jailbreak	Perturbation-based	×	✓
Dong et al. (2023)	VLMs	Jailbreak	Perturbation-based	×	✓
Rando et al. (2024)	VLMs	Jailbreak	Perturbation-based	×	✓
Cheng et al. (2024)	VLMs	Jailbreak	Perturbation-based	×	✓
Wang et al. (2024)	VLMs	Jailbreak	Perturbation-based	✓	×
Zero-Jack (Chen et al. 2024)	VLMs	Jailbreak	Perturbation-based	×	✓
Shayegani et al. (2023)	VLMs	Jailbreak	Embedding-based	×	✓
Gong et al. (2023)	VLMs	Jailbreak	Embedding-based	✓	×
Ma et al. (2024)	VLMs	Jailbreak	Embedding-based	✓	×
Wang et al. (2024)	VLMs	Jailbreak	Embedding-based	✓	×
SAVJ (Zhang et al. 2024)	VLMs	Jailbreak	Embedding-based	✓	×
Zou et al. (2024)	VLMs	Jailbreak	Other attack	✓	×
Tao et al. (2024)	VLMs	Jailbreak	Other attack	×	✓
Wang et al. (2024)	VLMs	Jailbreak	Other attack	✓	×
Zhang et al. (2024)	VLMs	Jailbreak	Other attack	✓	×
Liu et al. (2024)	VLMs	Jailbreak	Other attack	✓	×
Liu et al. (2024)	VLMs	Jailbreak	Other attack	✓	×
CS-DJ (Yang et al. 2025)	VLMs	Jailbreak	Other attack	✓	×
SI-Attack (Zhao et al. 2025)	VLMs	Jailbreak	Other attack	✓	×
HIMRD (Teng et al. 2024)	VLMs	Jailbreak	Other attack	✓	×
Gao et al. (2023)	Text-to-Image	Adversarial attack	Character-level	×	✓
Zhuang et al. (2023)	Text-to-Image	Adversarial attack	Character-level	✓	×
Du et al. (2023)	Text-to-Image	Adversarial attack	Word-level	×	✓
Maus et al. (2023)	Text-to-Image	Adversarial attack	Word-level	✓	×
Shahgir et al. (2023)	Text-to-Image	Jailbreak	Word-level	✓	×
Milliere (2022)	Text-to-Image	Jailbreak	Word-level	✓	×
JMA (Ma et al. 2024)	Text-to-Image	Jailbreak	Word-level	✓	×
Gao et al. (2024)	Text-to-Image	Jailbreak	Word-level	✓	×
Ring-A-Bell (Tsai et al. 2023)	Text-to-Image	Adversarial attack	Sentence-level	×	✓
Liu et al. (2023)	Text-to-Image	Adversarial attack	Sentence-level	✓	×
Yang et al. (2024)	Text-to-Image	Adversarial attack	Sentence-level	×	✓
Groot (Liu et al. 2024)	Text-to-Image	Jailbreak	Sentence-level	✓	×

the generation probability of a targeted harmful response $R = \{y_1, y_2, ..., y_n\}$. Analogous to jailbreak techniques in LLMs, this is generally formulated as minimizing the negative log-likelihood:

$$\min_{x_{adv}} -\sum_{i=1}^{n} \log p(y_i|t, x_{adv}) \tag{11}$$

where p denotes the model's next-token prediction probability.

Early works primarily address this optimization in white-box settings using gradient descent. Carlini et al. (2024) pioneered this by applying PGD directly to image pixels, optimizing from random noise to maximize the likelihood of harmful responses. Building upon this, Qi et al. (2024)

introduced semantic preservation constraints, optimizing existing benign images against a few-shot toxic corpus. Extending the attack surface to dual modalities, Wang et al. (2024) proposed jointly optimizing both random image noise and adversarial text suffixes to further elevate the attack success rate.

To systematically categorize visual prompt attacks, Niu et al. (2024) proposed distinct frameworks based on the attacker's capabilities: generating adversarial visual prompts from scratch (imgJP) or adding bounded perturbations to initial images (deltaJP). Furthermore, to address the critical challenge of cross-model transferability, both Niu et al. (2024) and Dong et al. (2023) explored ensemble-based strategies using surrogate models. Specifically, Dong et al. incorporated strict L_∞ norm constraints during optimization and employed spectrum simulation algorithms to ensure robust transferability to black-box targets.

As emerging multimodal architectures increasingly employ non-differentiable tokenizers or restrict access to model weights, traditional backpropagation becomes infeasible. To overcome non-differentiability, Rando et al. (2024) introduced continuous approximations for tokenizer shortcuts, enabling standard gradient-based updates for images. For strict black-box scenarios, Cheng et al. (2024) utilized surrogate models for a two-step dual-modal iterative refinement. Alternatively, Chen et al. (2024) proposed Zer0-Jack, which entirely circumvents the need for gradient access by leveraging zeroth-order optimization techniques.

4.1.2 Embedding-based attacks

Unlike perturbation-based attacks that inject imperceptible noise to manipulate the autoregressive generation, embedding-based attacks operate by directly embedding harmful semantics into the visual modality. This paradigm exploits the vulnerability of multimodal fusion, circumventing text-centric safety filters by injecting malicious intent either into the latent representation space or directly into the visual pixels. Let $\mathcal{I}(\cdot)$ denote the visual encoder and x the input image; the fundamental objective is to craft an adversarial image x_{adv} such that its encoded feature $\mathcal{I}(x_{adv})$ inherently conveys the targeted malicious semantics.

Latent space alignment From a representation learning perspective, Shayegani et al. (2023) demonstrated that benign-appearing images can be optimized to activate harmful regions within the VLM's latent space. Instead of directly targeting the generation loss, they formulate a compositional attack that minimizes the ℓ_2 distance between the image embedding $\mathcal{I}(x_{adv})$ and predefined malicious anchors (e.g., embeddings of harmful text, OCR triggers, or toxic visual features). By pairing this semantically toxic image with a vaguely worded benign prompt (e.g., "teach me how to make this stuff"), the attack effectively bypasses alignment guardrails without requiring explicit harmful text.

Typographical and visual role-play attacks A more direct, gradient-free manifestation of this paradigm relies on typographical embedding, where adversaries exploit the VLM's Optical Character Recognition (OCR) capabilities. Since most alignment efforts heavily prioritize text-based inputs, rendering malicious instructions as images effectively blinds the safety filters. Gong et al. (2023) pioneered this with FigStep, a black-box method that rephrases prohibited queries into declarative typographical images, pairing them with benign incitement prompts to trigger step-by-step completion.

To enhance the contextual deception and generalizability of typographic attacks, subsequent research has integrated visual role-playing and obfuscation techniques. Ma et al. (2024) introduced Visual Role-Play (VRP), a framework that utilizes Stable Diffusion to generate images of high-risk characters, concatenating them with typographic trait descriptions and queries. This plunges the VLM into a visually induced malicious context, overpowering its safety constraints. Similarly, Wang et al. (2024) proposed MultiModal Linkage (MML), which combines typographical rendering with word replacement (obfuscation), effectively guiding the model to autonomously decrypt and execute the concealed intent during inference. Furthermore, SAVJ (Zhang et al. 2024) generalizes this typographic transformation into a unified black-box framework, demonstrating high efficiency and transferability across various multimodal architectures.

4.1.3 Other methods

In addition to perturbation-based and embedding-based methods, there are some approaches that cannot be easily categorized. Zou et al. (2024) demonstrated that a logic-based flowchart can effectively jailbreak VLMs and, for the first time, present Flow-JD, a novel dataset specifically designed to assess the jailbreak capabilities of logic-based flowcharts in VLMs. Tao et al. (2024) proposed ImgTrojan by adopting a data-poisoning approach. They contaminate the training dataset by injecting poisoned (image, text) pairs. These pairs replace the original textual captions with malicious jailbreak prompts. During training for the image captioning task, they optimize the probability of generating a caption y given an image x_{img} and an instruction x_{des} for describing the image, denoted as $P_\theta(y \mid x_{\text{des}}, x_{\text{img}})$. In contrast to ImgTrojan, Wang et al. (2024) proposed a novel jailbreak method named IDEATOR. The method employs a VLM to generate targeted jailbreak texts, which are paired with jailbreak images created by a state-of-the-art diffusion model. Experiments demonstrate that it achieves a

high success rate in advanced VLMs. Unlike IDEATOR, which relies solely on a VLM, Liu et al. (2024) proposed a framework that implements an automated multimodal jailbreak attack. In this framework, visual jailbreak prompts are generated by a red team VLM, while textual prompts are crafted by a red team LLM guided by a reinforcement learning agent. Zhang et al. (2024) discovered that hidden "meta-instructions" influence how the model interprets images and steer its outputs toward generating harmful content. Gu et al. (2024) demonstrated that the adversary can simply jailbreak a single multimodal LLM agent, and without any further intervention from the adversary, all multimodal LLM agents will become infected exponentially fast and exhibit harmful behaviors. Yang et al. (2025) proposed the Distraction Hypothesis, followed by a novel framework called Contrasting Subimage Distraction Jailbreaking (CS-DJ), to achieve jailbreaking by disrupting multimodal LLMs alignment through multi-level distraction strategies. Zhao et al. (2025) uncovered a critical phenomenon termed Shuffle Inconsistency, which reveals a misalignment between multimodal LLMs' comprehension capabilities and their safety mechanisms when processing shuffled harmful instructions. Leveraging this discovery, they introduce SI-Attack, a novel text-image jailbreak attack framework. This method strategically selects the most harmful shuffled inputs by utilizing feedback from a toxic judge model. Teng et al. (2024) proposed a heuristic-induced multimodal risk distribution jailbreak attack method, called HIMRD, which consists of two elements: multimodal risk distribution strategy and a heuristic-induced search strategy.

The existing jailbreak attacks uncover specific vulnerabilities in VLMs, underscoring the need for improved safety mechanisms. These attacks highlight the critical gaps in current security frameworks, which could potentially be exploited to manipulate the model's behavior or bypass ethical safeguards. Therefore, there is a pressing need for the development of more robust defenses and safety protocols to mitigate such risks and enhance the reliability of VLMs.

4.2 Adversarial attacks on text-to-image models

Text-to-image (T2I) diffusion models, as a prominent paradigm of multimodal generation, have exhibited extraordinary capabilities in synthesizing high-fidelity images (e.g., Stable Diffusion). However, their widespread adoption has catalyzed rigorous investigations into their robustness against adversarial exploitation.

In this section, we systematically review adversarial attacks on T2I models. Based on the granularity of the applied perturbations, we categorize these methodologies into three distinct levels: (1) character-level, (2) word-level, and (3) sentence-level attacks.

4.2.1 Character-level attacks

Character-level attacks manipulate outputs by introducing imperceptible typographical alterations to specific characters within the original prompt (as illustrated in Fig. 10). Generally, the objective is to find an adversarial sequence c' that maximizes the divergence $\mathcal{D}$ between the clean generation distribution $p_\theta(x|c)$ and the adversarial generation $p_\theta(x|c')$, subject to a bounded edit distance $d(c, c') \leq \epsilon$:

$$\max_{c'} \mathcal{D}(p_\theta(x|c') \parallel p_\theta(x|c)), \quad s.t.\ d(c, c') \leq \epsilon \tag{12}$$

To quantify this divergence, Gao et al. (2023) proposed utilizing the Maximum Mean Discrepancy (MMD). Their method identifies crucial keywords in the clean prompt and applies non-gradient greedy search to introduce typos, glyph alterations, or phonetic shifts. Rather than directly attacking the diffusion process, Zhuang et al. (2023) demonstrated that generating character-level suffixes to minimize the cosine similarity in the CLIP text-encoder space is highly effective. They further extended this to targeted attacks by projecting the embedding divergence onto specific semantic directions, successfully manipulating the output without requiring white-box access to the diffusion model.

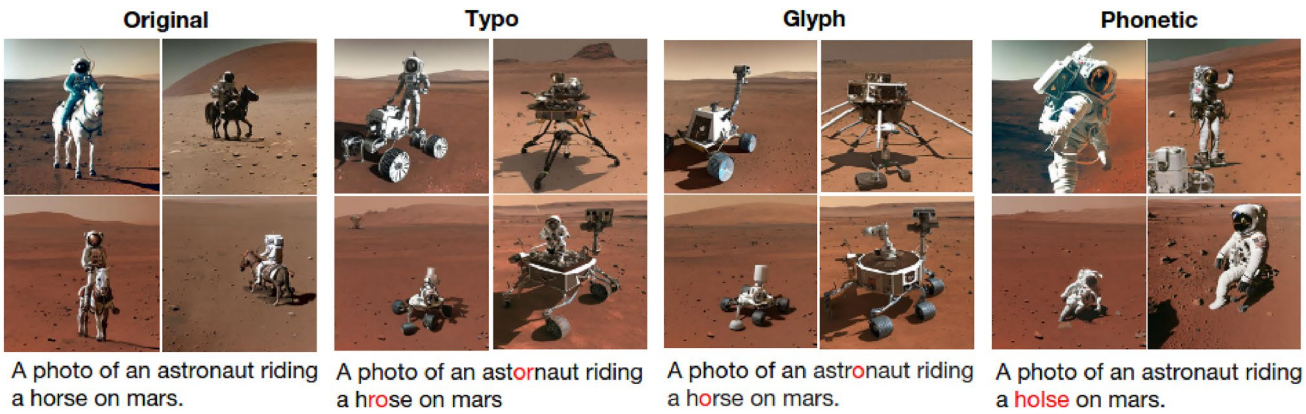


Fig. 10 An illustration of Gao et al.'s character-level attack method against Stable Diffusion (2023)

4.2.2 Word-level attacks

While character-level perturbations are effective, they often disrupt tokenization and are vulnerable to spell-checkers. Word-level attacks address this by substituting or optimizing entire tokens to induce semantic drift while preserving syntactic structure. Du et al. (2023) proposed ATM, a white-box gradient-based approach that maximizes a classification loss on the generated image to deviate from the original semantics, tightly regularized by fluency and semantic similarity constraints. Conversely, for black-box environments, Maus et al. (2023) developed a gradient-free pipeline that optimizes candidate word embeddings using surrogate losses.

In the context of bypassing safety filters (jailbreaking), word-level perturbations must navigate a complex decision boundary. Yang et al. (2024) introduced SneakyPrompt, which leverages reinforcement learning to guide token substitutions, successfully crossing safety boundaries while maintaining semantic dissimilarity from explicit triggers. Furthermore, Milliere (2022) revealed that utilizing macaronic or evocative prompting can naturally bypass safety checks. Recent advancements by Ma et al. (2024) and Gao et al. (2024) further formalize this by employing continuous embedding space optimization (JPA) and heuristic token search algorithms, respectively Fig 11.

4.2.3 Sentence-level attacks

Despite their success, both character- and word-level attacks often result in high perplexity (PPL), making them easily detectable by post-hoc language models. Sentence-level attacks mitigate this by transforming entire prompts to ensure readability, naturalness, and semantic coherence.

Tsai et al. (2023) introduced Ring-A-Bell, a conceptual red-teaming framework that extracts an empirical representation of targeted sensitive concepts in the latent space. It then injects this concept vector into the continuous representation of a benign prompt and decodes it back into a discrete adversarial sentence. In contrast, Liu et al. (2023) formulated the optimization to strictly decouple image similarity from text distance. By utilizing genetic algorithms, their framework generates adversarial sentences that are semantically distant from the original prompt but yield highly similar visual outputs.

For jailbreak-specific tasks, attackers exploit multimodal reasoning gaps. Yang et al. (2024) proposed the MMA attack, leveraging both textual and visual modalities iteratively to bypass safety guidelines. Similarly, Groot (Liu et al. 2024) automates the jailbreak process through tree-based semantic transformations, ensuring the adversarial sentences remain logically sound yet deceptive to safety filters.

Ultimately, the progression from character-level typos to sophisticated sentence-level semantic transformations highlights the fragility of current T2I models. Despite the deployment of prompt filters and post-generation safety checkers, frameworks like MMA (Yang et al. 2024) and SneakyPrompt (Yang et al. 2024) persistently demonstrate that adversarial robustness in multimodal generation remains an open and critical challenge.

5 Adversarial vulnerabilities in LLM-integrated ecosystems

As LLMs transition from isolated text generators to central reasoning engines within pervasive systems, their integration with external components, ranging from Web APIs and physical actuators to SQL databases, has introduced complex attack surfaces. These vulnerabilities primarily stem from the unverified trust established between the LLM core and its interacting peripherals, rather than intrinsic flaws within the model itself.



Fig. 11 An illustration of a word-level jailbreak attack on text-to-image models (Yang et al. 2024)

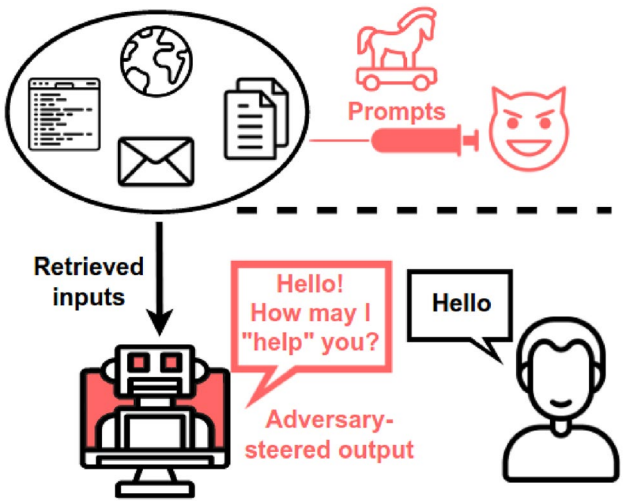


Fig. 12 An illustration of an indirect prompt injection attack on LLM-integrated applications. Attackers inject harmful content into the knowledge base, which is then retrieved by the LLM and incorporated into its generation process, leading to the production of harmful outputs (Greshake et al. 2023)

Table 3 A list of the surveyed works in LLM-Integrated system attacks

Work	Victim model	Attack strategy	Attacker knowledge	
			Black-box	White-box
Greshake et al. (2023)	LLM-Integrated Application	Prompt injection	✓	×
Pedro et al. (2023)	LLM-Integrated Application	Prompt injection	✓	×
Liu et al. (2023)	LLM-Integrated Application	Prompt injection	✓	×
Robey et al. (2025)	LLM-Integrated Application	Prompt injection	✓	×
Lu et al. (2025)	LLM-Integrated Application	Gradient-based	✓	×
Zou et al. (2024)	RAG	Prompt injection	✓	×
Chaudhari et al. (2024)	RAG	Gradient-based	×	✓

A critical manifestation of this systemic risk is the *indirect prompt injection* attack (see Fig. 12), where malicious payloads embedded within external data sources silently hijack the model's instruction-following logic. To systematically analyze these threats, we structure our discussion based on the architectural purpose of the integration. First, we examine Attacks on General LLM-Integrated Applications, which focus on general-purpose orchestration frameworks where LLMs actively invoke external tools, query structured databases, or manipulate physical environments. Second, we provide a dedicated analysis of Attacks on Retrieval-Augmented Generation (RAG) systems. While RAG is technically a specific subset of integrated applications, its unique reliance on unstructured knowledge pipelines and semantic vector spaces introduces specialized threat vectors (e.g., corpus poisoning and retrieval hijacking) that warrant a distinct, in-depth categorization. A synthesis of landmark research across this integrated ecosystem is provided in Table 3.

5.1 Attacks on LLM-integrated applications

LLM-integrated applications utilize language models to orchestrate complex tasks by connecting them with external environments. However, this architectural integration fundamentally blurs the strict boundary between "trusted user instructions" and "unverified external data". If an attacker injects malicious tokens into an external resource (e.g., a webpage or a user profile), the LLM may seamlessly parse this retrieved data as a high-priority instruction, executing the attacker's intent rather than the user's (Greshake et al. 2023). Depending on the scope of the integration, this systemic vulnerability propagates through two primary vectors: digital orchestration frameworks and physical intelligent systems.

Digital orchestration and textual injections Within digital software boundaries, LLMs are frequently deployed to orchestrate APIs, databases, and web tools, fundamentally shifting them from passive responders to active reasoning agents. Greshake et al. (2023) provided the foundational taxonomy for these indirect prompt injection threats in such applications, categorizing delivery mechanisms into passive, active, user-driven, and hidden injections. Expanding on this, Pedro et al. (2023) demonstrated how these conceptual vulnerabilities translate into severe traditional cyberattacks, introducing Prompt-to-SQL (P_2SQL) injection. By targeting orchestration frameworks like LangChain, P_2SQL poisons the retrieved database inputs, tricking the LLM into generating malicious SQL queries that can leak or destruct backend records. To systematize the exploitation of these applications, Liu et al. (2023) proposed HOU YI, a black-box prompt injection framework inspired by traditional web vulnerabilities. HOU YI employs a surrogate LLM to reverse-engineer the target application's hidden context via extensive input–output probing. Subsequently, it constructs and dynamically refines a structured injection payload to bypass framework-level defenses. The generation of such sophisticated payloads often requires navigating a complex discrete token-space to ensure the injected malicious instructions remain stealthy while maintaining high attack success rates against the aligned LLM core. As orchestration tools grant LLMs unprecedented read/write access to external environments, these optimized textual adversaries

become critical vectors for large-scale data exfiltration and privilege escalation.

Vulnerability propagation in physical intelligent systems Crucially, the scope of LLM-integrated applications extends beyond digital text processing into pervasive intelligent systems, such as embodied robotics and cyber-physical IoT ecosystems. In these physical applications, the LLM core is integrated with multimodal sensors and physical actuators, translating high-level cognition directly into physical actions. Consequently, vulnerabilities in the LLM reasoning process bypass digital security protocols and directly manipulate the system's actuators. To specifically illustrate this threat trajectory, Robey et al. (2025) introduced the first algorithmic jailbreak designed specifically for LLM-controlled robots. Their empirical study demonstrated that malicious prompts could successfully bypass cognitive guardrails in commercial systems, such as the Unitree Go2 robot dog and autonomous driving LLMs, causing them to execute harmful physical actions in the real world. From a mechanistic perspective, these vulnerabilities highlight a profound misalignment between textual safety training and physical execution boundaries. Expanding this threat into the multimodal perception layer, Lu et al. (2025) recently exposed the critical threat of physical-world adversarial patches on Vision-Language-Action (VLA) models. Instead of poisoning a digital database, an attacker can place a seamlessly crafted universal adversarial patch within a robot's physical workspace. When the VLA model processes this visual input, it acts as a physical-layer indirect injection. This visual perturbation actively misdirects the internal latent representations of the model, short-circuiting the textual safety mechanisms before they can evaluate the causal mapping of the generated physical action. This evolution demonstrates that as integrated applications expand from digital APIs to physical agents, the consequences of exploitation scale from data manipulation to tangible physical harm, demanding a deeper diagnostic framework for adversarial robustness.

5.2 Attacks on retrieval-augmented generation (RAG) systems

Retrieval-Augmented Generation (RAG) mitigates LLM hallucinations by dynamically fetching relevant external knowledge. However, the reliance on an external corpus introduces a critical data-poisoning attack surface. Unlike application-layer attacks that exploit the instruction-parsing phase, attacks on RAG systems focus on corrupting the retrieval pipeline itself.

Zou et al. (2024) formalized this vulnerability with PoisonedRAG, framing knowledge corruption as a joint optimization problem. The attacker crafts malicious texts designed to achieve two simultaneous goals: guaranteeing high similarity with the user's query (ensuring retrieval) and carrying specific semantic triggers that force the LLM to generate the attacker's desired incorrect response.

Chaudhari et al. (2024) advanced this concept by introducing Phantom, a white-box attack that compromises the RAG system by injecting a single, highly optimized adversarial document. Phantom employs a tripartite adversarial optimization strategy. First, it utilizes token-level gradients (e.g., HotFlip) to manipulate the document's embedding, maximizing its cosine similarity to specific user queries while hiding it from benign queries. Second, it incorporates adversarial suffixes–similar to the GCG attack–to forcefully break the LLM's alignment. Finally, it seamlessly embeds the exact malicious payload the attacker wishes to execute.

Summary The integration of LLMs with external orchestration components and retrieval databases undeniably amplifies system capabilities, yet it inherently introduces compounding security risks. As demonstrated by indirect prompt injections and RAG manipulation, attackers can effortlessly exploit the systemic trust placed in external data. While current defense strategies–such as input sanitization, strict role-prompting, and database access controls–provide foundational security, they struggle to contain the full spectrum of adaptive adversarial techniques. Future research must urgently pivot towards robust multi-agent verification, explainable retrieval filtering, and strictly sandboxed LLM execution environments to ensure the secure deployment of integrated AI systems.

6 Challenges and future directions for secure LLM-driven pervasive intelligent systems

Despite the burgeoning research on Large Language Model (LLM) vulnerabilities, adversarial exploitations in this domain encounter significant hurdles that impede the development of resilient human–machine interfaces. These limitations not only hinder the synthesis of potent adversarial samples but also obscure our understanding of the fundamental weaknesses within integrated AI ecosystems. This section explores three core dimensions of adversarial research and systemic defense: stealthiness, transferability, and explainability. It also presents concrete research agendas to guide future work.

6.1 Stealthiness and human-perceptual imperceptibility

Ensuring the stealthiness of adversarial modifications is a paramount challenge for both unimodal and multimodal

interaction security. In the context of human–machine collaboration, stealthiness refers to the subtlety of adversarial perturbations that remain undetectable to human users while successfully manipulating the model's behavioral trajectory. Current methodologies, however, struggle to balance attack potency with linguistic or visual naturalness. For instance, the Greedy Coordinate Gradient (GCG) attack often yields nonsensical token sequences that are easily flagged by perplexity-based filters. Conversely, while frameworks like AutoDAN improve sentence fluency to bypass statistical defenses, they frequently lack the semantic coherence required for authentic human-like interaction, diminishing their efficacy in practical, high-stakes scenarios.

6.2 Cross-model transferability and universal robustness

The transferability of adversarial inputs remains a formidable barrier to the realization of black-box exploitations. A lack of generalization often renders an adversarial sample effective against a specific architecture while failing against others, even those performing identical tasks. This inconsistency severely limits the threat model in pervasive computing environments where target parameters are typically inaccessible. Although recent advancements like the COLD attack and Tree of Attacks (TAP) have demonstrated improved cross-model efficacy, they have yet to achieve the status of universal attacks capable of generalizing across diverse model families. Developing robust, universal adversarial strategies remains essential for identifying systemic blind spots without prior architectural knowledge.

6.3 Explainability and transparent defense mechanisms

Explainable adversarial research is vital for fostering transparency and trust in human–machine systems. By providing insights into how adversarial effects propagate through neural layers, explainability enables the engineering of more interpretable and robust defense mechanisms. Current research, however, remains largely opaque, with many methodologies relying on auxiliary LLMs to generate adversarial variants, which further obscures the underlying causal mechanics. This lack of interpretability not only restricts the secure deployment of LLMs in sensitive domains but also hinders the development of semantic-aware defenses. Bridging this gap is critical for transitioning from empirical observation to a theoretically grounded understanding of adversarial robustness in intelligent systems.

6.4 Concrete future research agendas

To overcome the theoretical challenges of stealthiness, transferability, and explainability, the community must pivot from exploiting isolated vulnerabilities toward engineering systemic, proactive security architectures. Catalyzing this transition within pervasive intelligent systems requires executing a multifaceted research agenda. First, to move beyond the limitations of empirical red-teaming, future research must advance formal verification methodologies tailored for agentic orchestration. By drawing inspiration from structural proof theory and linear logic, frameworks can model API interactions as bounded resources, mathematically guaranteeing safe state transitions even against optimized adversarial prompts. Second, as deployments in localized IoT environments introduce critical latency constraints where cloud-centric security filters fail, it is imperative to engineer dynamic, asynchronous monitoring systems at the edge. This entails optimizing high-concurrency stream evaluations to intercept stealthy token streams in real-time without introducing system bottlenecks. Concurrently, to mitigate physical-layer multimodal injections, architectures must establish a mechanistic causal mapping for cross-modal consistency, ensuring that any physical actuation is structurally vetoed if it lacks a causally consistent textual justification. Furthermore, because universally transferable attacks will inevitably bypass single-model guardrails, systemic resilience demands distributed trust via multi-agent protocols. Implementing Byzantine fault-tolerant consensus mechanisms among independent critic agents allows systems to collectively sanitize inputs and isolate poisoned payloads before execution. Finally, evaluating these systemic robustness properties necessitates a paradigm shift toward continuous and pervasive security benchmarks; the field must move beyond static text corpora to develop dynamic, physics-simulated suites capable of assessing real-time adversarial perturbations and spatial safety violations in embodied environments.

7 Conclusion

While the exceptional capabilities of Large Language Models (LLMs) have significantly advanced the efficiency of human–machine interactions, their susceptibility to adversarial exploitations raises profound concerns regarding their reliability in critical, pervasive applications. A wide spectrum of attack methodologies has emerged, designed to craft adversarial examples that circumvent safety alignments and induce malicious behaviors. In this survey, we have systematically reviewed the landscape of adversarial attacks on LLMs, structuring our analysis around the evolution of their

interaction architectures: unimodal interfaces, multimodal systems, and LLM-integrated ecosystems.

For each architectural paradigm, we provided a detailed taxonomy and synthesized significant research. Specifically, within unimodal interactions, we examined the mechanics of jailbreak and prompt injection attacks that dismantle textual safety guardrails. For multimodal systems, we explored how cross-modal dependencies in vision-language models and text-to-image systems introduce complex vulnerabilities. In the context of LLM-integrated systems, we analyzed threats targeting LLM-powered applications and RAG frameworks, highlighting risks born from unverified trust between cognitive engines and external components. Finally, we analyzed the persistent challenges of current adversarial methods, such as stealthiness, transferability, and explainability–and outlined future directions. Ultimately, transitioning from empirical vulnerability discovery to the engineering of transparent defenses is imperative for fostering calibrated trust in next-generation human–machine systems.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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